StratumRO Execution Plan: Cadastre and PUG in QGIS first, then SNAP/ESA and CAD

Prepared 21 October 2026 for the StratumRO-QGIS project. This is a separate companion to the "StratumRO Multimodal Architecture Report". That report answers which models to use. This one answers what to do, in what order, and how to avoid drifting off the project.

Evidence labels: **[REPO]** read in your repository, **[PUB]** from a cited public source, **[CALC]** calculated from cited numbers, **[EST]** my estimate or recommendation, **[UNVERIFIED]** I could not confirm it.

Execution line

1. Direct answers to your questions

Did the first report include ESA SNAP after the PUG and QGIS phase? Only in one line. The first report listed SNAP and NDVI as a later tool adapter (stage 8 of its roadmap) and nothing more. It had no entry criteria, no data plan and no checks. This report fixes that: SNAP is Phase 4, it starts only after a defined exit gate (Gate A) for the QGIS phase, and section 8 says exactly what to do in it.

What other aspects should you cover? Ten, listed in section 10: data rights, backup, legal wording, tests, documentation, agent rules, cost tracking, data acquisition for parcels and intravilan, error taxonomy, and a stable environment. Several of them matter more right now than any new model.

How do you avoid deviating? Use the phase order, the gate checklists, and the scope-lock table in section 11. The repo already contains the same discipline (an `AGENTS.md` rule file, evaluation gates, a no-automatic-approval rule). The plan reuses it instead of inventing a second system.

2. What the project is, according to your repo

- StratumRO coordinates the surveyor's work across data, tools and specialised applications, including QGIS, and its current priority is the cadastral flow, with PUG and urbanism analysis next **[REPO]** (`docs/PROJECT_STATE.md`, `AGENTS.md`).
- QGIS is Phase 1 of the product. CAD engines (AutoCAD, Civil 3D, TopoLT) are delivery targets through open DXF and `.cp` coordinate tables in Stereo 70 (EPSG:3844) **[REPO]**.
- Non-negotiable rules in `AGENTS.md`: all cadastral computation in EPSG:3844, no invented coordinates, no approval without an authorized surveyor under ANCPI Order 600/2023, distinct physical objects stay distinct, no speculative rewrites, no new cloud service or Rust rewrite without an approved requirement **[REPO]**.
- Validated area is a small sample of Cluj-Napoca, about 0.420 km² (67 of 144 windows with valid RGB data). The platform is described as experimental **[REPO]**.
- Cluj measurements for orientation: recall 6.15% (4 of 65 references) and matched mean IoU 69.5%, but the references are legal parcels, not roofs **[REPO]**.

This gives the project line: **reliable cadastral roof and building geometry, then a PUG base map, both in QGIS, both signed off by a human, then other software and data sources.** Everything below serves that line.

3. Phase 0: Freeze and clean (do this first)

Nothing here needs a model. Each item removes a source of confusion that would otherwise contaminate later results.

#	Task	Why	Evidence
0.1	Confirm the exact GPUs and RAM on both machines	The repo's machine shows an RTX 4050 Laptop with 6,141 MiB, and NVIDIA does not list an "RTX 4090 Super". All VRAM plans depend on this	**[REPO]**, see architecture report
0.2	Write one data-facts sheet: source orthophoto step 1.1651 cm, processing cache 15 cm, viewing window about 3.9 cm per pixel, LiDAR density, CRS, vertical datum	The repo itself says the raster step does not prove real photo resolution or positional accuracy, and a cache at 15 cm is about 12.9 times coarser than the source	**[REPO]** `docs/STRATUMRO_ORTHO_RESOLUTION_NODATA_2026-10-17.md`
0.3	Settle the toolchain facts	CloudCompare is listed as not installed. SNAP 11.0.0 is called operational in one report and not verified in another, because the audit could not run the Java application	**[REPO]** `reports/cluj/08_open_source_toolchain_inventory.md`, `KILO_INDEPENDENT_PHASE2_AUDIT.md`
0.4	Keep Python environments separate	The project venv is Python 3.10.10 and QGIS ships Python 3.12.7. SAM 3's requirements list Python 3.12 or newer (its README badge says 3.10/3.11, so check), which would call for its own environment	**[REPO]**, **[PUB]** SAM 3 requirements as in the architecture report
0.5	Update stale legal text in code and docs	`stratum_ro/pug_product.py` cites Law 350/2001, which the new Code repealed (see section 6)	**[REPO]**, **[PUB]**
0.6	Set data-rights and cloud rules (section 10.1)	Needed before any survey data goes to a cloud model	**[EST]**

#	Task	Why	Evidence
0.7	Freeze the repo conventions: run `git status -s` first, one writer per file, no merge to `main` without approval, run the three verification commands before handoff	These are already your rules	[REPO] `AGENTS.md`

4. Phase 1: Evaluation base (before any training or model swap)

The repo's own evaluation-gates document says test lots must be reviewed, hashed and spatially separate, and that a passing preflight does not prove correct extraction [REPO] (`docs/STRATUMRO_EVALUATION_GATES.md`). Build on that.

1. **Frozen test area.** A spatially separate Cluj area that is never used for training or tuning. Record its boundary, tile hashes and creation date.
2. **Roof-level references.** Label the visible roof outline, keep instance IDs, preserve courtyards, never mark unlabeled pixels as background, include verified negatives (pavement, parking, trees, shadows), and record GSD, CRS and annotation version [REPO] (your own research note).
3. **Separate roof from ground footprint.** The current reference set is legal parcels, so recall against it understates quality and cannot measure roof extraction [REPO].
4. **Metrics to compute:** object precision and recall, boundary distance in metres, merge and split errors, count of invalid geometries, and operator correction time per building [EST].
5. **Thresholds set by you and the surveyor, written down before the first run.** I am not proposing numbers. The thresholds in `data/ground_truth/README.md` (for example boundary RMSE of 0.10 m for the ANCPI gate and 0.30 m for the PUG gate) are labelled there as inspired by Orders 600/2023 and 904/2023. The Order 600/2023 traceability note in the same repo says it found no exact article that supports its numeric cutoffs [REPO], and the text of Order 904/2023 that I retrieved states no accuracy requirements (section 6). Treat those numbers as project targets, not legal limits [UNVERIFIED].
6. **Initial labels.** About 200 to 300 corrected roofs is my working figure for a first trained model [EST]; the corrected SAM drafts are the cheapest way to produce them.

Exit of Phase 1: a hashed manifest of the frozen area, the reference layer, the metric scripts, and a baseline result for the current pipeline on it.

5. Phase 2: Cadastre in QGIS

5.1 Current state

- The reviewed-candidate workflow produces an `OBSERVED_ROOF_HYPOTHESIS`, with `ground_footprint=None` and `ground_status=NOT_MEASURED`, and the verdict stays `PENDING_OWNER_REVIEW`. The SAM 2 score is not cadastral accuracy **[REPO]** (`docs/STRATUMRO_CADASTRAL_WORKFLOW_INTEGRATION_2026-10-17.md`).
- Earlier regional work applied a 0.40 m eave setback to 195 buildings, while the newer cadastral route deliberately does not apply it automatically **[REPO]** (`docs/PROJECT_STATE.md`). These two policies conflict.
- The 19 October audit found 59 of 216 inspected results with visible errors, and 58 cases left incomplete and unrecovered **[REPO]**.
- ANCPI Order 600/2023, Art. 37(1), supports only general technical reception checks. The numeric thresholds in code (45 m^2 and 8 m^2, 5 degrees, 0.5 m edge) are project heuristics, and one docstring disagrees with the code on the edge length **[REPO]** (`docs/ANCPI_ORDIN_600_TRACEABILITY.md`).

5.2 Tasks in order

1. Decide the roof-to-ground policy once, with the surveyor: is the deliverable a roof outline, a ground footprint, or both, and is an eave setback applied, with what value and measured how. Record it in one decision file and apply it in one place **[EST]**.
2. Resolve the 0.5 m versus 1.0 m edge check and the MultiPolygon edge loop that is skipped, then add tests **[REPO]**.
3. Run the baseline (current SAM 2 tiny pipeline) on the frozen area and record the numbers.
4. Run the model A/B from the architecture report on the same area: SAM 2.1, the 4-channel RGB plus nDSM segmenter, and optionally a frozen DINOv3 head. Adopt a change only if it beats the baseline on the frozen area.
5. Replace the VLM verifier on the delivery path, or fix it, as described in the architecture report: its defaults approve missing fields at 0.90 confidence **[REPO]**.
6. Fix the NoData contract (mask kept through mosaic, crop, export; unknown pixels ignored in training) before any training run **[REPO]** (`docs/STRATUMRO_ORTHO_RESOLUTION_NODATA_2026-10-17.md`).
7. Export DXF with TopoLT/ANCPI-style layers and `.cp` coordinate tables in EPSG:3844, and re-open them in QGIS to check coordinates, CRS and layer names **[REPO]**.
8. Surveyor review and stamp. Nothing is "validated" before that **[REPO]**.

6. Phase 3: PUG base in QGIS

6.1 The legal ground has changed

- Law 169/2026, the Code of Territorial Planning, Urbanism and Construction, was published in the Official Gazette no. 661 on 10 August 2026 and entered into force on 25 August 2026 (Juridice.ro, AICPS) **[PUB]**.
- It repeals, among others, Law 350/2001 on territorial planning and urbanism and Law 50/1991 on construction authorization (BPV Grigorescu) **[PUB]**.
- PUGs older than 7 years must be updated within 3 years of entry into force, and existing PUGs stay valid until then (Art. 573, Sintact) **[PUB]**.

- Procedures started before entry into force stay under the earlier rules (Art. 582(1)), and after entry into force local authorities start new PUG documentation in GIS format with the additional content the Code requires (Art. 582(3)) (Sintact) **[PUB]**.
- The Code says the PUG is produced in a GIS with vector data for regulated entities (administrative limits, parcel limits, functional zones, buildings, roads, networks) plus attribute data, and that all planning documentation is prepared in a standardized GIS structure for publication in the national urbanism geoportal (Code text, CGRBV) **[PUB]**.
- Practical dates: the Code obliges holders of such data to send it to the ministry in GIS format within 90 days of entry into force, which I calculate as about 23 November 2026 **[PUB, CALC]**; authorities must revise their procedures by 31 December 2026 (Art. 578) **[PUB]**.
- New standard forms were approved by MDLPA Orders 975/2026 (construction authorization) and 1,022/2026 (urbanism documentation), according to a municipal notice (Târgu Mureș) **[PUB]**; I did not read the orders themselves **[UNVERIFIED]**.

What I could not verify: whether Order 233/2016 (cited in your handoff for PUG content) and Law 24/2007 on green spaces (cited in the repo's PUG module) remain in force in the same form under the Code **[UNVERIFIED]**. Check both against the current consolidated text before writing any PUG-related wording.

6.2 The data format rule you can implement now

The technical norms on spatial datasets for urbanism documentation (Order 904 of 15 May 2023, Official Gazette no. 470 of 29 May 2023) say, according to the Legislative Portal text **[PUB]**:

Requirement	Article
Datasets in GeoPackage, OGC GPKG version 1.3.1 or higher	Art. 7
Stereo 70, EPSG:3844	Art. 2
Vector objects only as simple polygon, line or point (ISO 19125-1)	Art. 8
Functional zoning: no overlaps, no gaps; for PUG, no gaps between zoning and the administrative boundary	Art. 9(1)
Networks connected by nodes; no overlapping segments within one object; points must not overlap	Art. 9(2)-(3)
Standardized folder and file structure; names with underscores and no diacritics	Art. 4(1)-(2)
Unencrypted ZIP archive to the ministry	Art. 4(3)-(4)
Zoning vectorized at parcel level where possible	Art. 10(2)

Feature-class names visible in the text I retrieved: `PlanSpatial`, `ZFExistenta`, `ZFPropusa`, `ZonaReglementareSuplimentara`, and the non-spatial `RegulamentLocalUrbanism`. The full schema is reproduced as a facsimile image that the text did not include, so take the exact field list from the official order and from the MDLPA working templates (version 1.1, 28 June 2024, CC BY 4.0, datelocale.mdlpa.ro) **[PUB]**. The text I retrieved states no scale, accuracy or metadata requirement **[PUB]**. Whether this order still applies unchanged after the Code is **[UNVERIFIED]**.

6.3 What your repo has and lacks

- The UAT Iepurești base is open data: buildings from OSM with a +0.50 m buffer, roads as OSM axes buffered 3 m (marked a technical approximation, not official footprints), and a UAT limit from OSM that is not an official ANCPI boundary **[REPO]** (`docs/HANDOFF_UAT_IEPURESTI_PUG.md`).

- The 8 October QA report lists 140 buildings, 45 road polygons, 81 forest patches and 20 ponds, all valid, with `imobile_cadastrale` at zero features as a declared data gap (the ANCPI WFS was unavailable). It shows `limita_intravilan` with 1 feature, while the earlier handoff had it at 0. Find the source of that intravilan polygon before using it **[REPO]**.
- The handoff states that a full PUG needs written parts, a multicriteria diagnosis, approved zoning, an official intravilan boundary, supporting studies, public consultation, approvals from authorities, and a local-council decision, and that none of this can be generated from OSM plus AI **[REPO]**.
- `pug_product.py` computes LOD1 volumes, a green-space register and POT/CUT per UTR cell **[REPO]**. POT and CUT need parcel or UTR polygons, and parcels are a data gap, so those indicators cannot be treated as real until the inputs are official **[EST]**.

6.4 Tasks in order

1. Name the product correctly: "existing-situation base map / working document, not an approved PUG". Keep the QGIS layer-group rule that proposals and restrictions stay empty, or carry an explicit "conceptual scenario, not approved" label **[REPO]**.
2. Get the official inputs: parcels and intravilan boundary through an official channel. I did not research the ANCPI access procedure **[UNVERIFIED]**.
3. Rebuild the layer tree to the Order 904/2023 structure: GeoPackage, EPSG:3844, simple geometries, feature-class names and fields from the official templates.
4. Add automatic checks for the Art. 9 topology rules (no overlaps, no gaps against the boundary) in the QA step, and run them headless **[EST]**.
5. Use LiDAR for what it measures well: heights, LOD1 volumes and green volume. State point density and class sources in the provenance log (Cluj is about 7.2 points per m^2 **[CALC]**).
6. Produce the provenance table (source, extraction date, URL, known limits) per layer, as the handoff already requires **[REPO]**.
7. Hand off to a qualified planner. A PUG is signed off by certified specialists, not by the pipeline **[EST]**, consistent with the repo's rule that no output is approved without the responsible professional.

7. Gate A: exit criteria for the QGIS phase

Phase 4 and Phase 5 do not start until all of these are true. Thresholds you must fill in are marked.

#	Criterion	Check
A1	Frozen test area, hashed, with roof-level references	Manifest and hashes exist
A2	Baseline and best candidate measured on it	Object precision and recall, boundary distance in metres, merge/split errors recorded
A3	Quality thresholds met	Thresholds written in Phase 1 by you and the surveyor: ____
A4	Roof-to-ground policy decided and applied in one place	Decision file and tests
A5	All outputs in EPSG:3844, zero invalid geometries	`validate_project.py --mode fast` and geometry report
A6	DXF and `.cp` exports re-open in QGIS with correct coordinates	Reopen check
A7	PUG base GeoPackage passes the Art. 9 topology checks, gaps declared, labelled as working document	QA report

#	Criterion	Check
A8	Surveyor review recorded, with verdict states kept separate (pending, rejected, accepted)	Owner verdict file
A9	Legal text refreshed to the current Code, with unverified items listed	Updated docs
A10	The three repo verification commands pass	Command output saved

8. Phase 4: SNAP / ESA (after Gate A)

8.1 Why it comes after, and what it is for

Your repo already says the same: SNAP is preserved for the later satellite and multispectral phase, because adding it to the RGB plus LiDAR pipeline would add Java overhead without a resolution benefit **[REPO]** (`reports/cluj/08_open_source_toolchain_inventory.md`). The scale gap is the reason. Sentinel-2 bands B02, B03, B04 and B08 are 10 m, and the L2A scene classification layer is 20 m (Copernicus documentation) **[PUB]**. One 10 m pixel covers 100 m^2, more than twice the 45 m^2 main-building threshold in your code **[CALC]**. So satellite NDVI gives context (vegetation, green-space indicators, change over time), and it does not delineate buildings or parcels.

8.2 What you already have

- For UAT Iepurești the repo holds four 10 m band clips (B02, B03, B04, B08, 763 by 1,101 pixels, EPSG:32635) from the Sentinel-2 L2A scene of 17 August 2026, tile T35TMK, with checks passed and attribution recorded **[REPO]** (`projects/UAT_Iepuresti/qa/03_raster_validation.md`).
- The clips do not include the scene classification layer. For cloud, shadow and snow masking add the SCL band **[REPO, PUB]**.
- Reflectance in L2A products carries a scale and an offset in the metadata (0.0001 and -0.1 in the product record I checked), so apply them before computing a ratio index (CDSE STAC) **[PUB]**. Confirm the values for your product's processing baseline.

8.3 Tasks in order

1. Verify the tool: run `gpt` and the SNAP GUI on this machine. Decide whether to stay on 11.0.0 or upgrade. The newest release notes I found are for SNAP 14.0.0, which lists hyperspectral and spectral-library tools, GDAL 3.12.1, STAC handling and several Sentinel-2 reader and Graph Builder fixes; the page states no release date, so check the download page (ESA STEP) **[PUB, UNVERIFIED]**.
2. Do the first NDVI in QGIS or GDAL as the reference. NDVI is (B08 - B04) / (B08 + B04) with reflectance values after scale and offset **[standard definition]**. For one date and one UAT clip, QGIS is enough. This gives a ground truth to compare SNAP with.
3. Rebuild the same NDVI as a SNAP Graph Builder graph and run it with `gpt`. Check that pixel values match the QGIS result within rounding. Only then use SNAP for what QGIS does not do well: Sentinel-1 SAR processing and repeatable batch graphs over time series **[REPO, EST]**.
4. Add the SCL cloud mask and, for time series, a quality rule such as a maximum masked fraction per AOI **[EST]**.
5. Reproject for combination with cadastral data. Keep EPSG:3844 for cadastral vectors and store the raster's native CRS and resampling method in the metadata, as the repo's NoData note requires for orthophotos **[REPO]**.

6. Produce layers for the PUG base: vegetation index, change maps, green-space context. Label them "context, 10 m". Cross-check with the ExG vegetation filter from the 1 cm orthophoto, since the two have very different resolution and timing **[EST]**.
7. Keep Sentinel attribution ("Contains modified Copernicus Sentinel data [year]") in every derived product **[REPO]**.

Do not use SNAP on the 1 cm orthophotos or on LiDAR; it is an Earth-observation tool.

9. Phase 5: CAD and point-cloud software (after Gate A)

Target	What to hand over	Checks	Evidence
AutoCAD / Civil 3D	DXF from StratumRO; LandXML for surfaces	Assign the coordinate system first; LandXML surfaces import directly	**[REPO]**, Autodesk help, drone deliverables guide **[PUB]**
Civil 3D point clouds	LAS, or LAZ in 2023 and later; ground-classified points for surfaces	Use classified clouds; a surface from the whole cloud can exceed the point limit	drone deliverables guide, Eagle Point **[PUB]**
CloudCompare	LAZ tiles for inspection and cloud-to-cloud comparison	Not installed on the repo machine; install and record the version first	**[REPO]**

Civil 3D keeps surfaces without an overflow file below roughly 1.5 to 2 million TIN points, so thin or tile the surface before building it (Eagle Point) **[PUB]**. The same source recommends classified clouds to avoid spikes **[PUB]**. Each target is an export adapter. It receives validated geometry and does not change how the geometry is made.

10. Other aspects you should cover

1. **Data rights and privacy.** Write down who owns and may share the Cluj orthophoto and LiDAR and any survey data, then classify each dataset as cloud-allowed or local-only. Nothing in the repo that I read states the licence for the Cluj imagery and LiDAR **[UNVERIFIED]**. Local Qwen is the fallback for local-only data. Recommendation **[EST]**.
2. **Backup and versioning.** Keep dataset and checkpoint hashes, a manifest per run, and an explicit storage decision for large files. The repo's policy says not to claim Git LFS is configured unless it is **[REPO]**.
3. **Legal wording.** Use "working document", "pending review" and "context" consistently. Never use "validated", "ANCPI-compliant" or "PUG" for unreviewed output **[REPO]**.
4. **Tests as gates.** Run `python scripts/validate_project.py --mode fast`, `python -m unittest discover stratum_ro/test` and `python tools/verify_ground_truth_and_metrics.py` before every handoff **[REPO]**.
5. **Documentation consolidation.** The repo has 251 Markdown files and about 0.61M tokens of docs **[REPO]**. Keep one current state file, one rule file, and one decision log; archive the rest. This also keeps the AI context pack small (12.6K tokens today).
6. **Agent rules for the hybrid workflow.** Every agent (Claude, Gemini, OpenCode models, local Qwen) reads `AGENTS.md` first. One writer per file. A change is reported as status, isolated changes, test results, risks and decision needed **[REPO]**. Check each plan's current usage limits before you rely on it for a long job; I did not research the limits.
7. **Cost and time log.** Record tokens, GPU hours and operator minutes per task. The architecture report shows a text-only agent step costs on the order of 1,150 tokens per candidate and a VLM step several hours of local time for 380 candidates, so the log shows where the budget goes **[CALC]**.

8. **Data acquisition plan.** List what is missing: parcels, official intravilan boundary, utilities, protection zones, and the PUG written parts. For each, name the official source and the owner of the request **[EST]**.
9. **Error taxonomy.** The repo found 214 defect cases in nine categories on 19 October (overlaps, roads, vegetation, cemetery, tanks, seams, truncated shapes, rotation errors, NoData) **[REPO]**. Keep that list, count each category on the frozen area, and aim each change at one category.
10. **Stable environments.** Pin versions of QGIS (3.40.0), GDAL (3.9.3), PDAL (2.8.1), Python (venv 3.10.10, QGIS 3.12.7) and the model checkpoints, and record them per run **[REPO]**.

11. Scope lock: how not to deviate

Item	Status now	Starts when
Cadastre roofs in QGIS	In scope	Now (Phases 0-2)
PUG existing-situation base in QGIS	In scope	After the Phase 2 data and policy decisions
Model A/B (SAM 2.1, 4-channel segmenter, DINOv3 head, polygon model)	In scope, conditional	Only after Phase 1
VLM verifier on the delivery path	Deferred, or reduced to an audit	Only if it beats the embedding probe on your labels
LLM fine-tuning	Deferred	After RAG, tool use and evaluation exist, and with hundreds to thousands of good examples
SNAP / NDVI / SAR	Deferred	After Gate A
Civil 3D, AutoCAD, CloudCompare adapters	Deferred	After Gate A
Desktop app, Rust rewrite, new cloud service	Deferred, and barred without approval	After Gate A, with owner approval
Producing an actual approved PUG	Out of scope	Not a pipeline output

Change control: before adding a new model or tool, write three lines: what weakness in the frozen-test results it targets, which metric will show the effect, and what you will stop doing if it fails. If you cannot fill in the first line, the work is a deviation.

12. Order of work and effort

Order	Work	Relative effort [EST]
1	Phase 0 clean-up and decisions	Small
2	Phase 1 frozen area and references	Medium to large (labeling)
3	Phase 2 baseline, policy decision, model A/B	Medium
4	Phase 3 PUG base to the Order 904/2023 structure	Medium
5	Gate A review	Small
6	Phase 4 SNAP and Phase 5 CAD adapters, in either order	Medium each

I give no calendar durations because they depend on how fast you can get references labeled and on data access that I could not check.

13. Open questions that change the plan

1. Which area is the primary target: Cluj (cadastre test) or UAT Iepurești (PUG base), or both in sequence?
2. What are the exact GPU models in both machines?
3. Who can accept or reject results (you, a licensed surveyor, a client), and what verdict states do they use?
4. What are the data-rights terms for the Cluj orthophoto, LiDAR and any survey data?
5. Is the first PUG deliverable for a real commission or for a demonstration?

Sources

Repository files read (paths inside the project): `AGENTS.md`, `docs/PROJECT_STATE.md`, `docs/STRATUMRO_EVALUATION_GATES.md`, `docs/HANDOFF_UAT_IEPURESTI_PUG.md`, `docs/ANCPI_ORDIN_600_TRACEABILITY.md`, `docs/STRATUMRO_CADASTRAL_WORKFLOW_INTEGRATION_2026-10-17.md`, `docs/STRATUMRO_ORTHO_RESOLUTION_NODATA_2026-10-17.md`, `reports/cluj/08_open_source_toolchain_inventory.md`, `reports/cluj/KILO_INDEPENDENT_PHASE2_AUDIT.md`, `projects/UAT_Iepuresti/qa/03_raster_validation.md`, `projects/UAT_Iepuresti/qa/05_qa_report.md`, `data/ground_truth/README.md`, `stratum_ro/pug_product.py`. Repository: StratumRO-QGIS.

Legal and planning:

- Law 169/2026, Chamber of Deputies
- Code text (CGRBV copy)
- Sintact, Part VIII transitional provisions
- Juridice.ro, AICPS, BPV Grigorescu
- Order 904/2023 technical norms, Legislative Portal
- MDLPA data structure model
- Târgu Mureș notice on new forms

Satellite and CAD:

- Copernicus Sentinel-2 L2A documentation, CDSE STAC collection
- SNAP 14.0.0 release notes
- Autodesk LandXML help, Drone survey deliverables guide, Eagle Point large datasets